RAG-Based Framework for Intelligent Banking Assistants Leveraging Structured Financial Data

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Abstract – **The paper presents a novel Retrieval-Augmented Generation (RAG) framework for intelligent banking assistants, integrating structured financial and regulatory data to improve accuracy and compliance in enterprise scenarios. A two-stage architecture RAG system is suggested. At the embedding stage advanced semantic chunking and metadata enrichment of the documents is utilized. At the retrieval and generation stage, a single input is used merging the original prompt, retrieved context, and generation instructions, and is passed to an LLM to generate the final, compliance-checked response. The suggested methodology is experimentally evaluated with structured Excel QA pairs and unstructured PDF policy documents. Manual expert evaluation confirm the effectiveness of combining RAG with structured data for complex banking knowledge tasks, highlighting practical considerations for real-world deployment.**

Keywords – **intelligent banking assistants; retrieval-augmented generation; structured data; enterprise knowledge management; regulatory compliance.**

I. Introduction

Currently, numerous financial institutions are actively integrating intelligent banking assistants into their operations. The primary focus of these integrations has typically been directed toward enhancing customer interactions. While this trend aligns with the rapid evolution of customer service technology, it may inadvertently increase employee workloads due to assistant responses that are insufficiently precise or contextually inaccurate. Given the rapid technological developments, a measured and strategically planned integration process is essential, progressively moving from less critical internal services toward more critical customer-facing functionalities.

The initial focus of this research is to understand and address internal operational challenges faced by bank employees. Bank personnel frequently manage extensive documentation encompassing normative regulations, procedural guidelines, and organizational best practices. Efficient retrieval and accurate interpretation of such documents are vital for consistent regulatory compliance and high-quality customer support.

The paper presents a Retrieval-Augmented Generation (RAG) framework specifically designed to enhance employee capabilities by facilitating efficient access to structured procedural and regulatory information. Incorporating structured datasets within an internal RAG architecture helps employees address diverse operational scenarios in alignment with established banking standards and regulations. At this stage, the research prioritizes non-sensitive internal documentation to ensure robust performance, accuracy, and regulatory compliance. Adopting this incremental approach aims to mitigate risks associated with data privacy and regulatory standards, creating a solid foundation for future integration with sensitive financial data and broader customer-facing applications. The following sections detail the conceptual architecture, preliminary findings, and strategic insights essential for effective deployment of structured data-driven intelligent banking assistants.

II. Related Works

RAG is a recent advancement in Natural Language Processing (NLP) that combines retrieval mechanisms with generative language models to produce accurate and contextually relevant responses [1]. Initially popularized by applications in question-answering systems [2] and dialogue generation [3], RAG leverages external knowledge bases or document collections to enhance response quality. Hybrid NLP systems integrating RAG effectively bridge the gap between retrieval and generative approaches, mitigating common shortcomings such as factual inaccuracies and limited contextual awareness [4].

Structured data integration into NLP has been primarily explored within fields requiring precise and interpretable results, such as healthcare [5], finance [6], and legal domains [7]. Structured datasets offer substantial advantages by enabling exact information retrieval, improving accuracy, and ensuring clarity in generated outputs. Previous studies highlight the efficacy of using structured databases in generative models, emphasizing the robustness gained through structured contextualization [8].

Conversational AI specifically in finance and banking has attracted significant interest due to its potential to streamline customer interactions and operational efficiency. Applications range from basic account inquiries and transaction processing to complex advisory services [9]. While many solutions rely on unstructured textual sources or general knowledge bases, there is growing acknowledgment of the need to incorporate structured financial data to enhance accuracy and contextual specificity.

Given the heavily regulated nature of the banking industry, regulatory compliance in AI implementations remains paramount. Recent literature underscores the importance of compliance-driven AI solutions, outlining rigorous standards for data privacy, interpretability, and auditability [10]. Regulatory-compliant AI systems are critical to managing risk, maintaining trust, and ensuring long-term viability within highly sensitive domains such as finance and banking.

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III. Problem Formulation

The presented study addresses the challenge of streamlining the workload within Human Resources (HR) departments through the development of intelligent banking assistants enhanced by structured internal documentation. The primary focus is on effectively leveraging RAG systems to assist employees in quickly obtaining accurate answers related to internal organizational policies, principles, and legal requirements.

One of the significant challenges in this context is that normative documents and regulatory texts are predominantly stored in unstructured formats, such as PDFs, which are not optimized for embedding or semantic search. The lack of structure makes it difficult to extract relevant content efficiently and accurately, especially when dealing with long-form documents that contain complex hierarchical information.

The input data of the RAG system comprise:

- Employee-generated queries expressed in natural language, typically regarding organizational rules, HR policies, legal compliance, and internal procedural details.
- Structured internal data, including detailed organizational guidelines, company-specific rules, processes, and normative regulations.

The expected output is accurate, contextually relevant, and coherent natural language responses provided by the banking assistant, addressing the specific employee query and minimizing the necessity for HR personnel intervention.

Several constraints for utilization of RGA on the defined problem can be outlined:

- Data Privacy: Strict containment of all structured organizational data within internal institutional infrastructure, without external data transfer or cloud-based processing.
- Regulatory Compliance: Responses must align with legal standards, company-specific guidelines, and internal policy frameworks.
- Real-time Inference: Ensuring minimal response latency to support efficient, immediate interaction and reduce delays in workflow.

The key objectives in implementing RAG-based framework for intelligent banking assistants leveraging structured financial data are defined as follows:

- Accuracy: Ensuring high reliability and correctness of generated responses.
- Relevance: Precisely matching the retrieved information to the employee's query context.
- Efficiency: Significantly reducing response times, thus improving employee productivity and HR efficiency.
- Compliance: Adherence to both organizational and legal frameworks, thus enhancing institutional trust and mitigating compliance risks.

By addressing these parameters, the research aims to demonstrate the practicality and effectiveness of structured data-driven intelligent assistants in significantly reducing HR workload.

IV. Framework Overview

A. System Architecture

The proposed RAG system follows a modular pipeline developed in Python, utilizing frameworks such as LangChain and Lama Index. The two-stage architecture of the RAG system, as shown in Fig. 1, comprises:

- Embedding: Source documents are split into meaningful chunks, enriched with custom metadata, and indexed as vectors in a vector database.
- Retrieval and generation: The user's prompt is embedded and used to query the vector database for the most relevant chunks. The system then merges the original prompt, retrieved context, and generation instructions into a single input, which is passed to an LLM to generate the final, compliance-checked response.

B. Structured Data Processing

Initial structured data include HR-provided question-answer pairs in Excel format, converted into JSON objects to ensure accurate embedding without interference between data points. Later, the system handles complex, unstructured PDF documents. Data normalization and schema alignment are performed to ensure effective retrieval. Entity linking and temporal data handling are minimized at the initial phase.

C. Retrieval Module

The retrieval module employs dense retrieval techniques leveraging various embedding models. Initially, models like Paraphrase Multilingual and Roberta were tested, with moderate results. Transitioning to Lama Index provides better management of metadata and retrieval accuracy. Further improvements involve semantic splitting of PDF data using advanced techniques like SemanticSplitter, significantly enhancing retrieval precision.

D. Augmentation Strategy

Retrieved entries are augmented by enriching metadata – keywords, generated questions, and chunk-specific titles created via the language model – to enhance retrieval precision. This strategy optimizes context window usage and significantly improves response accuracy.

E. Generation Module

The bgGPT language model [11] is conditioned on structured contexts from optimized embedding retrievals. Personalization strategies are not explicitly applied, however, the augmentation process ensures a high-quality context for accurate generation. Post-processing maintaines the clarity, consistency, and regulatory compliance of generated responses.

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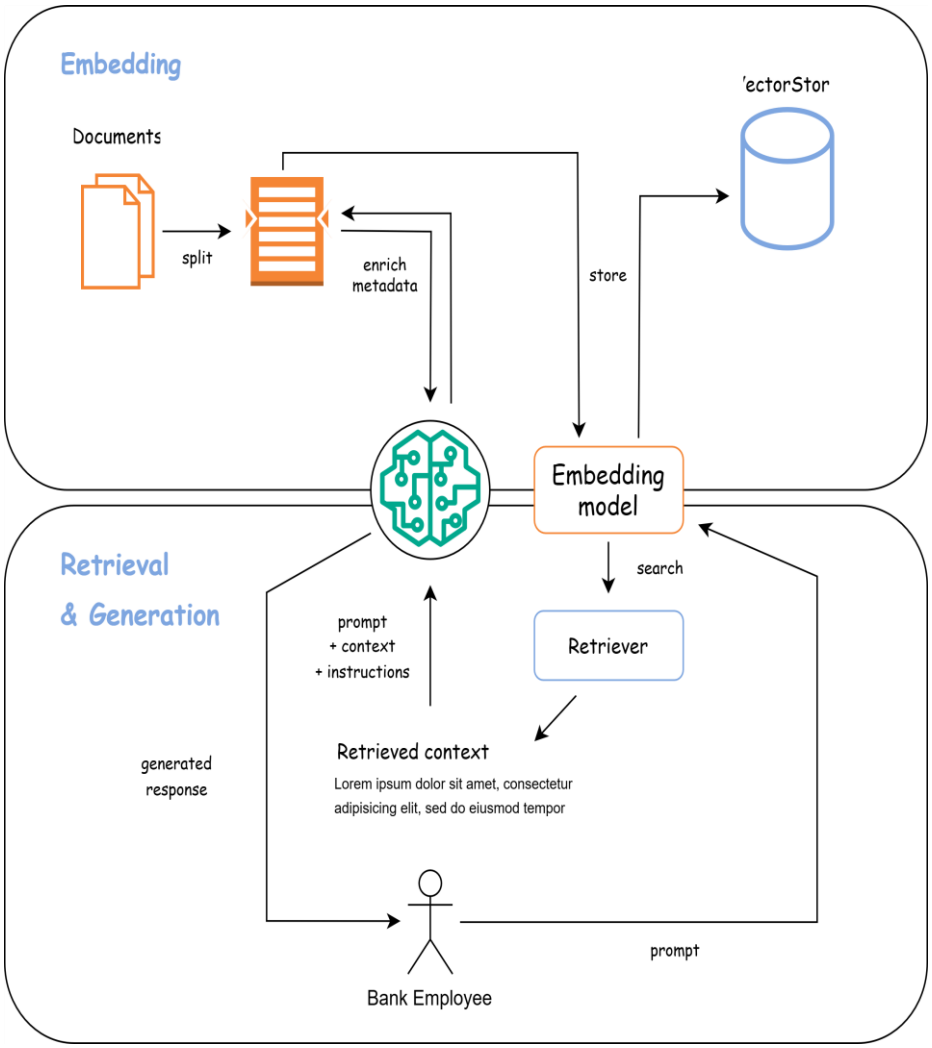


Fig. 1. System architecture of the proposed RAG system

V. Framework Implementation

The suggested RAG-based framework is implemented as part of an intelligent banking assistant in Bulgarian bank institution.

A. Phase 1: Excel Data Experiments

In Phase 1, the HR department provided a foundational dataset containing 100 common organizational question-answer pairs in Excel format. These pairs represent the most frequently encountered HR queries within the organization, aiming to streamline internal information retrieval and response processes. The experimental implementation employs Python frameworks, beginning with LangChain due to its straightforward integration and comprehensive toolset.

Initially, the embedding step involves evaluating embedding models such as Paraphrase Multilingual and Roberta. However, preliminary results indicated moderate effectiveness, primarily due to imprecise chunk embedding. Detailed analysis revealed that initial embedding inaccuracies arose from embedding multiple questions and answers within a single chunk, leading to confusion and irrelevant retrieval results.

To address this issue, a data transformation process is conducted, converting the Excel-based question-answer pairs into structured JSON format. Each JSON object distinctly represented one question-answer pair, ensuring isolated embedding and avoiding overlap between entries. This structural adjustment significantly enhances retrieval precision and consistency.

The subsequent shift from LangChain to Lama Index mark a strategic enhancement in framework capabilities, notably improving the control over embedding operations and metadata management. With Lama Index, precise control over vector database entries is feasible, allowing meticulous exclusion of default metadata that previously introduced noise into retrieval results. This targeted management substantially improves retrieval accuracy by approximately 15-20%, thereby achieving a satisfactory and stable system performance.

For response generation, the Bulgarian open-source bgGPT language model is utilized due to its linguistic accuracy and robust performance on Bulgarian-language inputs. The integration of this generative model successfully confirms the overall feasibility and reliability of the system, effectively responding to user queries in a contextually accurate and coherent manner.

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B. Phase 2: PDF Document Experiments

Phase 2 addresses the challenge posed by processing extensive, unstructured PDF documents provided by the HR department. These documents contain dense, complex text spanning multiple pages per answer, complicating the accurate retrieval of relevant information.

Various document splitting strategies are methodically tested:

- *SimpleNodeSplitter*: suitable for straightforward documents but insufficient for complex text.
- *RecursiveSplitter*: effective in some scenarios but limited by hierarchical inconsistencies.
- *HierarchicalSplitter*: optimal only when clear document structure existed, such as headings or clear divisions.
- *SemanticSplitter*: ultimately selected for its superior semantic coherence and precise handling of complex textual content.

The SemanticSplitter employs embedding-driven evaluations, grouping sentences based on semantic similarity rather than mere textual proximity. This strategy proves significantly more effective, accurately preserving contextual continuity and thereby substantially enhancing retrieval precision.

Further advancements in retrieval effectiveness involve a comprehensive metadata enrichment strategy. The reintroduction of essential metadata, specifically tailored for complex PDF documents, includes generating five key terms per chunk. Additionally, each chunk is enriched through the bgGPT model by creating five representative questions and appropriate titles. This sophisticated metadata enrichment facilitates more precise and contextually relevant retrieval from the vector database.

Embedding model selection is increasingly critical as documents demonstrate nuanced semantic variations. Initial models such as Roberta and Paraphrase Multilingual are limited by their lower-dimensional embeddings, insufficient for nuanced differentiation. The OpenAI ADA model, with its higher-dimensional embeddings, markedly improves retrieval accuracy. Subsequent evaluation of advanced, open-source embedding models, specifically BG-M3, Labce, and Multil-E5, indicates Multil-E5 as the most effective, offering embeddings of approximately 1400 dimensions. This dimensional enhancement significantly increases semantic differentiation capacity, crucial in the banking domain, notably improving accuracy and specificity.

This comprehensive approach, combining sophisticated semantic chunking techniques, advanced metadata management, and strategic embedding model selection, culminates in substantial enhancements in retrieval accuracy and overall system efficacy, ensuring robust and reliable intelligent assistant operations.

VI. Framework Evaluation and Experimental Results

A. Evaluation Methodology

At this stage of the research no automated evaluation pipeline is implemented. Instead, performance is assessed entirely through manual inspection by HR domain experts. The reviewers focus on the following qualitative criteria:

- *Chunk Quality*: Inspecting the top retrieved chunks to verify that they contain the information needed to answer the query (completeness, lack of noise, proper segmentation).
- *Answer Correctness*: Judging whether the assistant's final answer fully and accurately addresses the question in accordance with official HR guidelines.
- *Factual Consistency*: Ensuring that every statement in the generated answer could be traced back to the retrieved source text.
- *Latency (Observed)*: Informally timing end to end response latency to confirm the system remains within a ~2 s threshold for usability.
- *User Satisfaction*: Collecting subjective feedback from a small group of pilot users regarding clarity and usefulness of the answers.

No numerical scores such as Recall@K or MRR are computed as their design and automation are deferred to the next stage of the project and will be treated in a follow up study focused on evaluation methodology.

B. Baselines

To isolate and measure the contribution of each of the enhancements described in the previous section, namely semantic chunking (SemanticSplitter), rich metadata enrichment, and high dimensional embeddings (Multil E5), four baseline configurations are defined as summarized in table 1:

- **Baseline 1 (B1)**: Default RAG (Simple Splitter, No Metadata):
  - Fixed size sentence splitter;
  - No custom metadata (only default source tags);
  - Paraphrase Multilingual embeddings + bgGPT generator;
- **Baseline 2 (B2)**: RAG + Semantic Splitter (No Metadata):
  - SemanticSplitter for improved chunk boundaries;
  - No custom metadata;
  - Paraphrase Multilingual embeddings + bgGPT generator;
- **Baseline 3 (B3)**: RAG + Metadata Enrichment (Semantic Splitter):
  - SemanticSplitter for improved chunk boundaries;
  - Five keywords, five auto generated questions, and a title per chunk stored as metadata;
  - Paraphrase Multilingual embeddings + bgGPT generator;
- **Baseline 4 (B4)**: RAG + Metadata Enrichment + High Dimensional Embeddings (Semantic Splitter):
  - SemanticSplitter for improved chunk boundaries;
  - Same metadata enrichment as Baseline 3;
  - Multil E5 embeddings (≈ 1400 dimensions) + bgGPT generator;

The proposed RAG-base system is identical to Baseline 4, representing the production grade configuration against which the three simpler baselines are compared.

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TABLE 1. BASELINE SYSTEM PROPERTIES

System	B1	B2	B3	B4
RAG	Default RAG	Default RAG	Default RAG	Default RAG
Splitter	Simple Splitter	Semantic Splitter	Semantic Splitter	Semantic Splitter
Metadata	No custom metadata	No custom metadata	Metadata Enrichment	Metadata Enrichment
High dimensional embeddings	No high dimensional embeddings	No high dimensional embeddings	Paraphrase Multilingual embeddings	Multil E5 embeddings

C. Experimental Results

In the first phase (Excel data), the main challenges stems from the lack of clear chunking (multiple question-answer pairs placed in a single chunk) and from poorly structured or noisy metadata during embedding. This led to context mixing and frequent errors in retrieving relevant information. After reformatting the data to JSON (ensuring one question-answer pair per chunk) and removing unnecessary metadata, results improved by approximately 15–20%. Manual HR expert assessment found that about 60–65% of the answers are fully correct in the simplest baseline configuration.

The second phase works with PDF documents and requires more precise processing techniques. The SemanticSplitter enables the creation of more meaningful chunks, and metadata enrichment is introduced (keywords, auto-generated questions, and titles for each chunk). Using Paraphrase Multilingual embeddings, accuracy increases to approximately 68% (Baseline 2) and adding metadata further improved results to about 78% (Baseline 3). When a higher-dimensional embedding model (Multil-E5) is added, the proportion of fully correct answers reached ~85% (Baseline 4/Proposed system). All results are based on manual HR expert validation of 50 randomly sampled queries. A summary of the manual evaluation of baseline comparison is given in Table 2. Latency in all configurations remains below 2 seconds.

TABLE 2. MANUAL EVALUATION OF BASELINE COMPARISON

RAG system	Fully correct answers
B1	~60%
B2	~68%
B3	~78%
B4 (***Proposed system***)	~85%

As no automated scoring pipeline is in place, exact numerical scores are not computed. Instead, HR reviewers manually label each answer as correct, partially correct, or incorrect. Across 50 sampled queries, the proportion of fully correct answers increases markedly with each successive baseline. Reviewers also note a substantial reduction in unnecessary HR escalation as answer correctness improves. Some of the qualitative examples reveal improved differentiation between closely related queries (e.g., "debit card" vs. "health insurance card") and enhanced clarity and specificity in responses to complex policy questions.

Increased embedding dimensions significantly improves accuracy but marginally raises computational resource requirements. However, response latency remains acceptable for real-time operations.

The system occasionally struggles with highly ambiguous or insufficiently specified queries highlighting areas for improvement in query interpretation and clarification capabilities. Additionally, performance heavily depends on embedding quality and meticulous metadata enrichment.

VII. CONCLUSION

The presented study demonstrates a practical approach to building a RAG system for banking and HR assistant scenarios, focusing on structured and semi-structured data in Bulgarian. The research is executed in two main phases: an initial phase with structured Excel question-answer pairs, and a more complex phase with unstructured PDF documents. Key innovative contributions in the suggested RAG-based system include semantic chunking, custom metadata enrichment, and high-dimensional embedding models. The evaluation by manual expert review confirms a substantial increase in the proportion of fully correct answers, from around 60% in the default baseline to 85% in the final system. Latency remained suitable for real-time applications.

The results highlight the importance of data preprocessing, chunking strategy, and embedding quality in real-world enterprise knowledge assistants. A careful, iterative approach starting with less critical, internal use-cases before scaling to sensitive customer-facing applications is essential for minimizing risk and maximizing organizational value.

Such RAG-based systems can greatly reduce employee workload, increase information accessibility, and provide more consistent, policy-compliant responses within complex institutional environments. This approach is particularly relevant in domains with extensive regulatory requirements and rapidly evolving internal guidelines.

The research lays the groundwork for next-generation intelligent assistants in banking and enterprise settings, with clear pathways for extending functionality, scalability, and privacy in future deployments.

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Several research directions are open for further exploration:

• Multimodal RAG: Expanding the framework to support voice (audio transcripts) and images, paving the way for rich, multimodal customer service and internal knowledge assistants.

• Real-time constraints: Further optimizing model serving and retrieval latency for production-scale, live deployments.

• User Feedback Loops: Implementing systematic feedback mechanisms to capture user corrections, improve the assistant iteratively, and enable continuous learning.

• Automated Evaluation: Developing robust, automated metrics and evaluation pipelines to complement expert assessment and enable large-scale, objective benchmarking.

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